CACHET · NOTES, SEALED

Test & Quality Report

Production build · Lighthouse 13.5.0 · 30 September 2026

Next.js 16 · React 19 · Hono · PostgreSQL
Drizzle ORM · Tailwind CSS 4

93
Avg. mobile performance, public pages

100
Avg. desktop performance, public pages

100
Avg. accessibility, all pages

42/42
Automated unit tests passing

1 · Lighthouse — mobile

Emulated mid-range phone, slow 4G (150 ms RTT, 1.6 Mbps), 4× CPU slowdown. Each page was audited 3 times; the median run is shown (Lighthouse's recommended practice, scores vary run to run).

PAGE	PERF.	ACCESS.	BEST PR.	SEO	FCP	LCP	TBT	CLS	SPEED INDEX
Landing page `/` runs: 93 · 93 · 93	93	100	100	100	1.2 s	3.1 s	120 ms	0	1.2 s
Sign in `/login` runs: 92 · 94 · 95	94	100	100	100	1.1 s	2.9 s	140 ms	0	1.1 s
Create account `/register` runs: 92 · 94 · 95	94	100	100	100	1.1 s	2.9 s	120 ms	0	1.1 s
Help center `/help` runs: 85 · 91 · 91	91	100	100	100	1.1 s	3.3 s	140 ms	0	1.1 s
Terms of Service `/terms` runs: 93 · 94 · 94	94	100	100	100	1.1 s	3.0 s	110 ms	0	1.1 s
Privacy Policy `/privacy` runs: 89 · 93 · 94	93	100	100	100	1.1 s	3.0 s	160 ms	0	1.1 s
Security `/security` runs: 89 · 92 · 92	92	100	100	100	1.1 s	3.2 s	90 ms	0	1.2 s
Shared note (recipient view) `/share/‹token›` runs: 88 · 89 · 89	89	100	100	63	1.1 s	3.6 s	100 ms	0	1.4 s
Dashboard `/dashboard` signed in runs: 79 · 82 · 84	82	100	100	100	1.2 s	4.0 s	250 ms	0	2.0 s
All notes `/notes` signed in runs: 77 · 82 · 84	82	100	100	100	1.2 s	4.1 s	230 ms	0	1.3 s
Note editor `/notes/‹id›` signed in runs: 80 · 81 · 81	81	100	100	100	1.2 s	4.1 s	270 ms	0	1.2 s

2 · Lighthouse — desktop

Desktop preset (fast connection, no CPU slowdown).

PAGE	PERF.	ACCESS.	BEST PR.	SEO	FCP	LCP	TBT	CLS	SPEED INDEX
Landing page `/`	100	100	100	100	0.3 s	0.8 s	20 ms	0	0.7 s
Sign in `/login`	100	100	100	100	0.3 s	0.7 s	30 ms	0	0.5 s
Create account `/register`	100	100	100	100	0.3 s	0.7 s	0 ms	0	0.5 s
Help center `/help`	100	100	100	100	0.3 s	0.7 s	10 ms	0	0.5 s
Terms of Service `/terms`	100	100	100	100	0.3 s	0.7 s	10 ms	0	0.5 s
Privacy Policy `/privacy`	100	100	100	100	0.3 s	0.7 s	0 ms	0	0.4 s
Security `/security`	100	100	100	100	0.3 s	0.7 s	20 ms	0	0.4 s
Shared note (recipient view) `/share/‹token›`	99	100	100	63	0.3 s	0.9 s	10 ms	0	0.5 s
Dashboard `/dashboard` signed in	99	100	100	100	0.3 s	0.9 s	10 ms	0	0.8 s
All notes `/notes` signed in	99	100	100	100	0.3 s	1.0 s	20 ms	0	0.6 s
Note editor `/notes/‹id›` signed in	99	100	100	100	0.3 s	0.9 s	0 ms	0	0.7 s

3 · Notes on the results

> **Shared-note page SEO (63) is intentional.** Share links are private and secret, so the page sends `noindex, nofollow` (header and meta tag) and `robots.txt` disallows `/share/`. Lighthouse counts "page is blocked from indexing" as an SEO failure; for a secret link it is the correct, secure behaviour.

> **Signed-in pages** (dashboard, notes, editor) were audited with a logged-in session. Public tools such as PageSpeed Insights cannot log in, so they can only test the public pages. The signed-in pages carry more interactive code (editor, sharing controls, live counters), which costs a few points on the strict mobile test; on desktop they score in the 90s.

> **Test setup.** `next build` + `next start` on a Windows laptop, with the database on the same machine. This mirrors production, where the Vercel functions run in Singapore (`sin1`) next to the Neon database. Numbers from PageSpeed Insights on the live URL will differ slightly because they depend on the test machine and network.

4 · Automated tests & checks

Unit tests (Vitest) — 42 passed, 0 failed

- `tests/crypto.test.ts` — 9 tests
- `tests/features.test.ts` — 13 tests
- `tests/share-rules.test.ts` — 20 tests

Covers token and key generation, scrypt hashing, share-link status rules, validation, safe redirects, the sensitive-data advisor, TOTP two-factor codes (RFC 6238 test vectors), editor formatting and preferences.

Static checks

- TypeScript strict type check (tsc --noEmit): **0 errors**
- ESLint (next/core-web-vitals + TypeScript): **0 problems**
- npm audit (all dependencies): **0 known vulnerabilities**
- Production build (next build): **compiled successfully**
- Hydration check — 17 pages, 4× CPU, browser in a different time zone than the server: **0 console errors**

Concurrency test (`npm run test:race`) — real server, parallel requests

```
✓ one-time public link — 1 success(es), statuses
{"200":1,"410":24}
✓ one-time view count — view_count = 1
✓ one-time password link — 1 success(es), statuses
{"200":1,"410":4}
✓ time-based view count — 25 successful views, view_count = 25
✓ parallel brute force capped — statuses {"423":25} (423 = locked;
at most 5 guesses are ever checked per lock window)
✓ revoked link — status 410
```

Security headers (live response)

- `Content-Security-Policy: script-src 'self' 'nonce-…' 'strict-dynamic'; object-src 'none'; base-uri 'self'; frame-ancestors 'none'`
- `Strict-Transport-Security: max-age=63072000; includeSubDomains; preload`
- `X-Frame-Options: DENY`
- `X-Content-Type-Options: nosniff`
- `Cross-Origin-Opener-Policy: same-origin`
- `Share pages → Referrer-Policy: no-referrer; X-Robots-Tag: noindex, nofollow, noarchive; Cache-Control: no-store, max-age=0`
- `API → Cache-Control: no-store, max-age=0`

Full Lighthouse reports (median runs, session cookie removed) are included in `docs/lighthouse/`.